

SEP Emotion Classifier - Final Report

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Abstract

This report presents our general strategy for six-class facial emotion recognition under practical constraints. Because compute resources are limited and several datasets are small or heterogeneous, we rely on compact models instead of large architectures. The core idea is to improve input quality first through structured data preparation and strong augmentation. This reduces instability and helps the model learn clearer patterns from limited data. The following sections summarize the approach and its technical design.

1. Introduction

1.1. Context and relevance

Facial emotion recognition is important for human-centered AI systems, but practical deployments are often constrained by limited compute and low-resolution inputs. In this project, we therefore focus on a compact and resource-aware setup.

1.2. Problem statement

Our main challenge is to classify six emotions from limited and heterogeneous data while keeping the model small. Small datasets and domain differences increase overfitting risk, and weak input signals make feature learning unstable.

1.3. Goal and research question

The goal is to reach high classification quality despite limited resources. Our central research question is: *How can strong data preparation and augmentation enable high accuracy with small models under constrained resources?*

2. Method and Approach

Our approach follows one practical rule: with limited resources, data quality and training stability are primary levers. Therefore, we combine a compact model with strict preprocessing, synchronized augmentation, and modular feature fusion.

2.1. Data preparation pipeline

1) Combined FER2013 and AffectNet-8 training setup

Guiding idea: combine complementary datasets under one consistent training interface. We prepare FER2013 and YOLO-cropped AffectNet-8 in a shared format with aligned splits and class mapping, so both datasets can be used in one joint training pipeline.

2) Landmark metadata generation

Guiding idea: use facial landmarks to guide the model toward important facial points and to better handle faces that are not perfectly aligned. For each sample, facial landmark coordinates are extracted, scaled to target resolution, and normalized. The vectors are stored in `landmarks` so the model receives explicit spatial hints (e.g., around eyes, eyebrows, nose, and mouth) during geometry-appearance fusion.

3) Signal shaping

Guiding idea: pass cleaner, more informative structure to the model earlier. Before model ingestion, images can be transformed with fixed edge operators (Sobel, Roberts, Laplacian, or raw baseline). This creates a controlled preprocessing axis without changing classifier topology.

2.2. Geometry-consistent augmentation

Guiding idea: increase data diversity. When an image is transformed, landmark coordinates are transformed with the same geometric mapping. Supported operations are random resized crop, rotation, translation, perspective warp, affine shear, and horizontal flip. Perspective uses homography mapping in homogeneous coordinates; affine operations use explicit matrix transforms. After transformation, coordinates are clipped and re-normalized.

2.3. Imbalance-aware optimization

Guiding idea: limit majority-class dominance in uneven datasets. Class imbalance is addressed with two modes: class-balanced undersampling and inverse-frequency loss weighting. This balances gradient contributions across emotion classes without changing the core architecture.

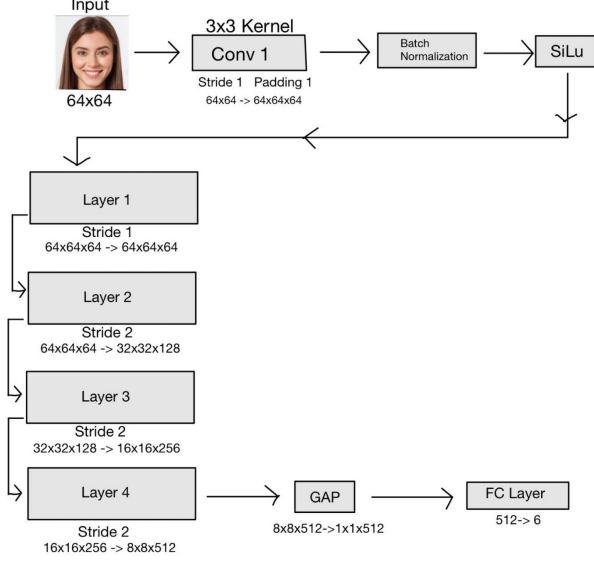


Figure 1. Pipeline overview of the classifier architecture. Starting from a 64×64 input image, the model applies an initial 3×3 convolution, four residual stages with stage-specific strides and feature-map dimensions, Global Average Pooling, and a final 6-node fully connected layer.

3. Model Architecture and Module Boundaries

3.1. A) Central emotion-classification model

For our model architecture we chose a modified ResNet18 Model. Instead of the traditional 224×224 image resolution our model accepts 64×64 images and to accommodate for that we made several architectural modifications. One of the main modifications was changing the initial convolution layer, as we replaced ResNet18’s 7×7 kernel with a 3×3 one to ensure that we do not lose critical information given the lower resolution input. The network then consists of 4 layers with each 2 blocks implementing residual stages. Each of the blocks use batch normalization followed by SiLU instead of ResNet’s classic ReLU, as the Swish activation function offers smoother gradients and slightly better optimization in deeper networks.

In addition, after the blocks we use a Global Average Pooling (GAP) instead of the traditional Flatten layer to reduce the total parameter count. Then we implemented a 6-node fully connected layer with a softmax function to finish our model off.

Optional internal branches of the central model **Raw landmark fusion.** Motivation: inject explicit facial geometry with minimal parameter overhead. Technical description: the landmark tensor is flattened to a fixed $2N$ vector and passed directly to late fusion without extra nonlinear layers. This preserves exact coordinate information and

avoids early compression artifacts. Integration: this branch is attached directly before the classifier head.

Multi-view heatmap branch. Motivation: represent landmark information at multiple spatial scales. Technical description: landmarks are projected into Gaussian heatmaps with multiple σ values:

$$H_k(x, y) = \sum_{i=1}^N \exp \left(-\frac{(x - x_i)^2 + (y - y_i)^2}{2\sigma_k^2 + \epsilon} \right).$$

Coordinates are clamped to the valid image grid before rasterization, then stacked maps are processed by a compact CNN (Conv-BN-SiLU blocks with global pooling) into a dense embedding. Integration: the pooled heatmap feature is appended to the head input.

Landmark-guided attention. Motivation: amplify expression-relevant regions before deep feature extraction. Technical description: image and heatmaps are fused to predict an attention mask $A(x, y)$ and apply

$$\tilde{I}(x, y) = I(x, y) \cdot (1 + A(x, y)).$$

The mask is produced by a small Conv-BN-SiLU stack followed by Sigmoid, so modulation remains bounded and stable. Integration: this modulation is applied in the main image path before deeper backbone stages, and an optional pooled feature from the fused tensor is also exposed to the classifier head.

Sobel feature branch. Motivation: preserve high-frequency edge structure as a complementary cue. Technical description: fixed Sobel kernels compute G_x and G_y , then gradient magnitude is formed as

$$M = \sqrt{G_x^2 + G_y^2 + \epsilon}.$$

Inputs are first remapped from normalized range to $[0, 1]$, and M is normalized per sample by its spatial maximum. The resulting edge map is encoded by a dedicated compact convolutional branch (Conv-BN-SiLU, global pooling). Integration: the resulting Sobel embedding is concatenated in the same head-level fusion vector.

Landmark GNN branch. Motivation: capture relational geometry between landmarks, not only absolute coordinates. Technical description: landmarks are treated as graph nodes with 2D coordinates, linearly projected to hidden embeddings, and updated for multiple message-passing rounds. Each round applies a shared message MLP, mean-aggregates node context, concatenates local and global context, and updates node states with a nonlinear projection. Final graph features are obtained through global pooling over nodes. Integration: the pooled graph embedding is injected into the classifier head with other branch features.

Dynamic head fusion. Motivation: enable strict ablations while keeping one unified model definition. Techni-

cal description: the FC input dimension is assembled dynamically from active components (backbone, raw landmarks, heatmap embedding, attention-derived feature, Sobel embedding, GNN embedding). Missing optional inputs are handled with zero-valued fallback tensors of matching shape, which keeps tensor dimensions deterministic across configurations. Integration: branch toggles change only this fusion composition, not the outer training pipeline.

3.2. B) Separate landmark-detector module

A dedicated landmark regressor is trained as an independent model. It uses direct coordinate regression ($2 \times \text{number of keypoints}$), affine+noise augmentation, and standard regression optimization. Its purpose is geometric keypoint estimation as a separate task module.

3.3. C) Separate face-cropper module

A dedicated face-cropper is trained as a separate multi-task model with two outputs: face confidence and normalized bounding box coordinates. It supports alternative backbones, boundary-aware bbox loss variants, and geometric/photometric augmentation. Its purpose is robust face localization as an upstream module.

3.4. D) Separate K-Means module (creative first idea)

An original idea was pure non-neural classification via K-Means as a baseline module. The concept is to use fixed feature spaces (flattened image features, optionally extended with landmarks), cluster into six groups, and map clusters to emotion labels.

4. Experiments

Metric focus. Training was performed over multiple epochs, because facial-expression patterns were not learned reliably in a single pass through the data. Model parameters were updated after each batch via backpropagation. In addition, a validation split was used to estimate generalization to unseen samples, which is critical in facial emotion recognition because models can learn dataset-specific patterns instead of expression-related features and thereby overfit. After each epoch, training accuracy and training loss were computed by comparing predicted and true labels.

Adam was selected as the primary optimizer due to its adaptive update mechanism and broad use in deep-learning practice. Initial tuning experiments evaluated learning rates between 0.0003 and 0.003, with repeated runs over multiple epochs.

To reduce overfitting, two regularization strategies were applied: Dropout and L_2 weight decay. Dropout was used to reduce reliance on individual neurons, and L_2 weight decay was introduced after overfitting was observed in early tests.

To identify a suitable trade-off between underfitting and overfitting, combinations of regularization settings and epoch counts were evaluated. More than 15 parameter combinations were tested repeatedly and averaged. Dropout rates in the range 0.1 to 0.5 were examined. For L_2 regularization, weight decay values of 10^{-3} , 10^{-4} , and 10^{-5} were compared.

Epoch counts were also varied. An initial configuration of 10 epochs showed underfitting. Further tests were conducted at 20 and 30 epochs, while runs above 30 epochs reached training accuracies above 99%, indicating overfitting. Batch size was additionally evaluated at 64, 128, and 256.

After introducing a custom transformation, 30 epochs were no longer sufficient, and additional epochs were required to avoid underfitting. Loss decreased more slowly with the custom transformation, but reached a comparable level after longer training, which motivated extending the epoch count. At this stage, evaluation functions for precision, recall, and F_1 -score were also integrated.

Furthermore, a landmark model and a new dataset were introduced. After renewed overfitting was observed, additional functions were implemented to improve test accuracy: a Sobel feature branch (separate CNN path for edge information) and width-multiplier support to simplify experiments with different image resolutions, which also influences regularization behavior implicitly.

Figure 2 shows qualitative predictions of the self-trained landmark detector on 9 samples (3×3 grid), including predicted outputs and labels. In this panel, landmark localization is visibly successful for most samples. Some samples are only moderately cropped or landmarked. For *Sample 36*, the facial landmarks are not detected.

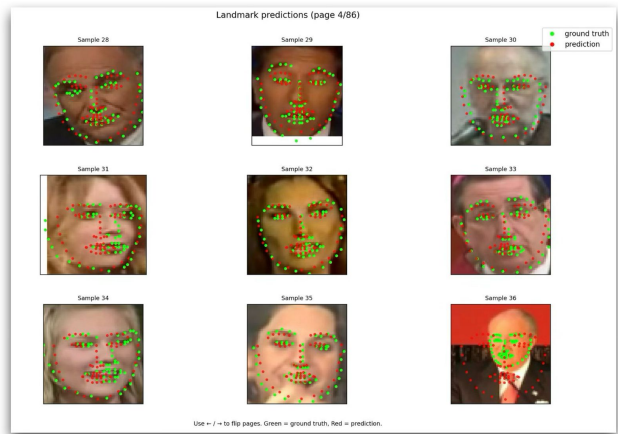


Figure 2. Qualitative landmark-detector results on 9 samples. Most faces show stable landmark localization, while *Sample 36* shows a clear miss.

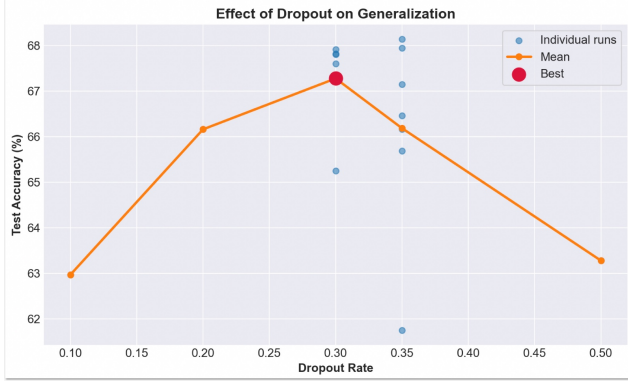


Figure 4. Effect of dropout rate on model generalization. The scatter plot and mean trend indicate peak performance around a dropout rate of 0.30, with a best recorded accuracy of approximately 67.3%.

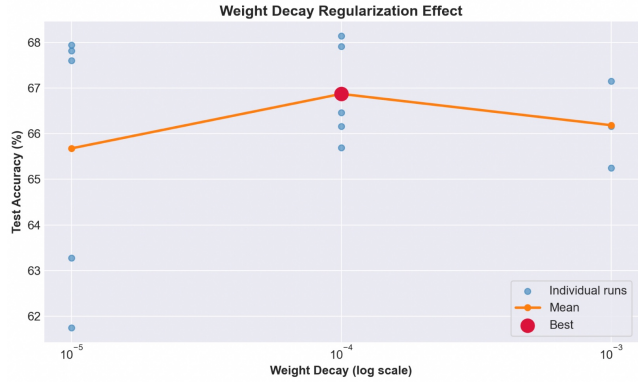


Figure 5. Impact of weight decay regularization on test accuracy. Performance is shown across weight-decay values on a logarithmic scale (10^{-5} to 10^{-3}); the best mean performance and highest individual run are observed at 10^{-4} .

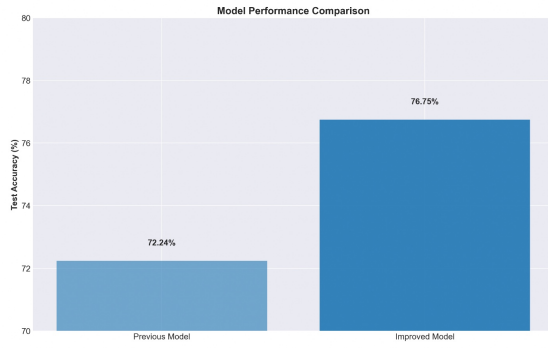


Figure 3. Model performance comparison between the initial architecture and the improved model. Test accuracy increases from 72.24% to 76.75%.

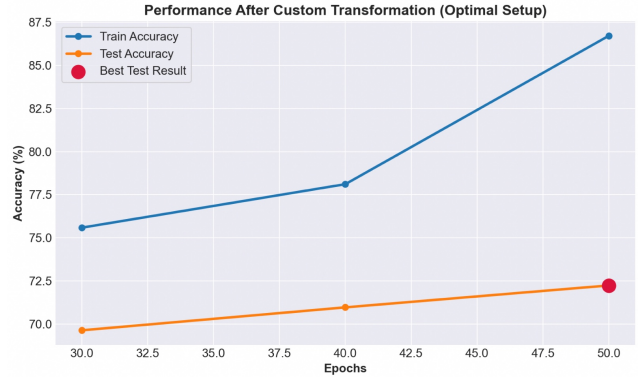


Figure 6. Training and test accuracy over epochs under the optimal setup after custom transformations. Training accuracy increases up to approximately 87%, while test accuracy improves more gradually and reaches 72.24% at 50 epochs.

5. Results

In early experiments, a learning rate of 0.001 produced the highest test accuracy. Other evaluated learning rates achieved test accuracies in the range of 64% to 66%, whereas 0.001 reached a maximum of 68%. Therefore, 0.001 was fixed for subsequent parameter studies.

For Dropout, the best-performing values were 0.30 and 0.35. Smaller Dropout values were associated with weak regularization and overfitting, while larger values led to underfitting because the network became too weak. Although 0.35 reached the highest single test accuracy (67.94%), 0.30 was selected for the final setup due to more consistent behavior and a better average test accuracy.

For weight decay, 10^{-3} performed best overall. Smaller values (10^{-4} , 10^{-5}) yielded lower test accuracy despite similar training accuracy, indicating overfitting; their penalty effect was too weak for robust regularization.

With respect to batch size, 64 improved generalization

but produced noisier gradients and slower training, whereas 256 reduced generalization performance. Consequently, 128 was selected as the final compromise.

Increasing training to 40 epochs raised accuracy to 70%. Extending training by an additional 10 epochs yielded a further 2.5% improvement, but also increased overfitting, reflected by a large training-test gap.

With the extended epoch schedule, F_1 -score improved from 0.59 to 0.69. Precision and recall remained similar, with no strong bias toward a specific error type. After integrating the landmark model and the additional helper functions, overall test accuracy increased by 4.5%, from 72.24% to 76.75%.

Best landmark-detector result: validation MAE = 4. Best face-cropper result: validation bbox MAE = 0.05. These MAE values refer to the validation set.

6. Discussion

A plausible limitation is data diversity: faces that are not perfectly aligned or that deviate strongly from the dominant framing appear less robustly handled by the landmark detector. This is consistent with failures like *Sample 36*, where landmarks are not recovered, while more regularly aligned faces are localized well. Since cropping and landmark quality are the foundation for the emotion classifier, we stopped further tuning of the self-trained detector/cropper setup. Instead, we switched to the prebuilt library (Bulat and Tzimiropoulos [1]) and used its outer landmark coordinates for cropping.

References

- [1] Adrian Bulat and Georgios Tzimiropoulos. How far are we from solving the 2d & 3d face alignment problem? (and a dataset of 230,000 3d facial landmarks). In *International Conference on Computer Vision*, 2017. 5